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A Concept of a Training Project IT Management System

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Abstract

The dynamic economic situation in the country and the world supports a high demand for training services. Their provision requires more and more advanced IT support. The purpose of this article is standardisation of the assumptions for the design of an IT system intended for managing business processes present within the framework of training implementation undertakings, including those carried out in the e-learning mode. It is assumed that the pre-project, conceptual modelling of business processes and operations necessary for the organization, running and settlement of training, developed by the authors, shall provide the designers with the revised domain knowledge and by this it will increase the quality of the proposed IT system supporting the functioning of a company providing training services. Also, it will increase the economic viability of the training projects managed via such a system.

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Keywords: data base, distance learning, management information system, training company, business process, concept modelling,

1. Introduction

The proposed process and architectural model of the IT system have been developed with the use of BPMN (Business Process Modeling Notation) technique based on business analysis typical of a training institution. The model was verified in pilot implementation projects including the implementation of the individual processes associated with the organisation of training at Technical University in Koszalin, comprising the use of the academic e-learning

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platform. The implementation and monitoring, in real-life conditions, of the key software modules used to assist the training management process has confirmed the compatibility of the proposed model with the adopted conceptual assumptions.

The studies carried out consist of:

- process modelling in BPML (Business Process Modeling Language),
- concept mapping,
- system modelling using UML (Universal Modeling Language).

2. Domain knowledge in IT projects

The term domain knowledge is used by IT professionals to denote true, verifiable knowledge relating to the defined area of human activity. When they use conceptual domain knowledge or refer to a domain expert opinion, they accentuate this way using the given specific domain which is the subject of discourse, interest or problem [1]. In the light of classification adopted in Poland, the knowledge necessary [2] to begin design works on training project management system belongs, generally, to the terminological (definitions), methodical (pragmatics), and normative (needs, expectations, criteria) areas of the domain.

In software engineering, the domain knowledge is the knowledge on the environment that the target IT system will be operating in [3]. Possessing an excellent knowledge of coding technique is not a value in itself. The project team may commence writing code only when they possess a good domain knowledge. That is, the team needs domain experts as much as technical experts.

Software developers usually acquire domain knowledge from the software users in the given domain (they are treated as domain experts). This knowledge usually concerns the workflow of the users, business data, economic policy, characteristic limitations, and key terms to software development. The domain knowledge acquired from the users is usually incomplete, informal and poorly organised. However, it should be transformed by the IT team designers into computer programs and databases.

Software development can even be defined as mapping domain knowledge onto the code, and therefore, mastering both coding and the domain is mandatory. One cannot write a system code without knowing what the system ought to do. In many cases, the software developer must understand the domain in an even more detailed way than a domain expert because the expert can solve all kinds of exceptions and cases relying on experience and common sense, using intuition, and the software developer must explicitly handle all those situations in the code before they arise.

The communication between software users and software developers is often difficult. They must create a common language to communicate. The development of common vocabulary, necessary for communication, may often take too long to fit the scheduled period of the project.

In untypical projects, as the one discussed herein, it is very difficult to obtain a domain expert. So, in order to provide advice on the construction of a training project management system, it is not enough to be an expert in education or in managing an enterprise, both specialisations Domain knowledge in IT projects

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In untypical projects, as the one discussed herein, it is very difficult to obtain a domain expert. So, in order to provide advice on the construction of a training project management system, it is not enough to be an expert in education or in managing an enterprise, both specialisations are necessary, and furthermore one must have advanced knowledge in the field of communication with the IT production team. The authors of this paper, due to their profession and experience in designing software for education, expect that the knowledge presented by them will turn out to be useful for training system creators.

3. Process approach to managing a training company

The assumption of the process approach to modelling of an IT management system is the optimisation of the company primarily in terms of process (not function that the software developers are usually used to). The purpose of the process approach is to obtain a high level of reliability in a specific business operation, here - in the organisation and conduct of training. This approach requires, first of all, to identify the business processes and to develop a map thereof [4]. The knowledge of the processes map allows for their improvement with the support from the persons managing the designed IT system.

Treating a typical training company as a process organization [5] is a kind of a simplification. In fact, training companies in Poland are not managed based on the process organisation concept. Nevertheless, due to the general trend of the popularisation of the process approach not only in the managerial environment but also among developers of IT management systems, the attempt to explore the potential of such an approach in a specific application appears to be justified.

When selecting the technique for the process analysis, the authors rejected the use of the robust, but less popular in the country, IDEF techniques IDEF [6] (Integration DEFinition Methods) for a more modern BPMN [7] (Business Process Modeling Notation). An important element in the preparation of process research has been the conceptual [8] analysis of the problem domain based on the ontological knowledge of the science of management [9]. The map of the main concept (terms) of the domain is illustrated in Fig. 1.

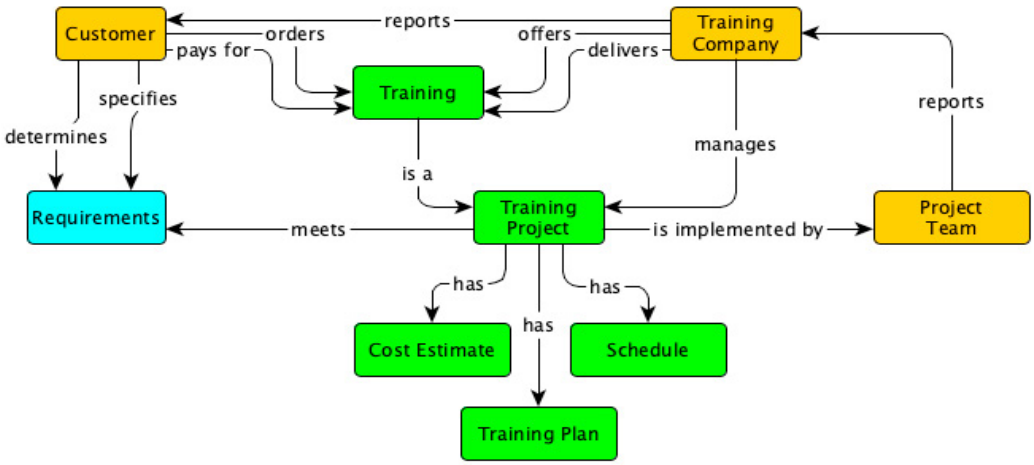

Fig. 1. The map of key concepts relating to a training undertaking (own study).

The key processes constituting the training management are:

- the process of managing the training offers (quoting process),
- the process of preparing the training contracts,
- the process of scheduling the training projects,
- the process of reporting and training settlement (see fig. 2). And so on

3.1. The process of managing the training offers

The company providing training receives many requests for quotation from various economic operators, including marketing agencies [10]. On the one hand, due to a tendering nature of the operation of agencies, only a small part of prepared bids reaches implementation. On the other hand, despite the costs arising from the preparation of the tenders, the bid-winning contracts are attractive both in terms of their substance and economically. The problem is that agencies often formulates only the idea of training and expect the contractor to provide technical and substantive advice aimed at clarification of the training implementation plan.

Also, in the case of other customers, the company making the settlement of the training must spend a lot of time on clarifying the former's requirements [11] - requests for quotation are usually very general. Even when the customer declares a training subject and scope that is well thought in detail, it usually lacks a detailed description of the training objectives, preliminary conditions, quality indicators and other elements of the tender necessary to make an offer. Sometimes, before preparing the quotation, the company providing the training must spend many hours on meetings with the customer to clarify the expectations.

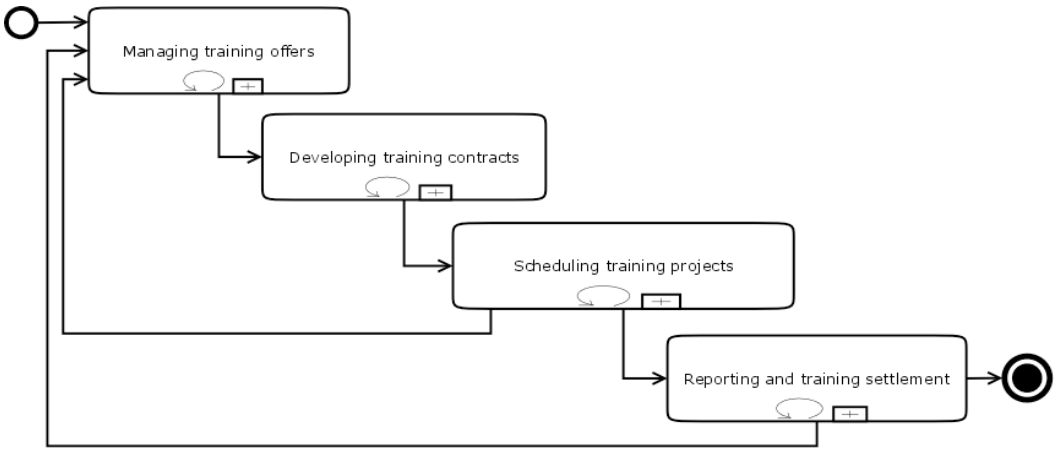

Fig. 2. The map of key business processes in a training undertaking (own study).

The training company may also mediate in the implementation of training - conclude contracts with external training providers, buy training materials from them, use the services of experts from external companies, under the signed cooperation agreements [12]. These more complex business configurations will further complicate the process of training management, and in the future - the scheduling, reporting, and training settlement processes.

It is also worth to notice the fact that the exact cost of the whole training project can be estimated only after obtaining a detailed hourly valuation of the training. After all, it's not an easy step in the quoting process. The training company employees are paid various hourly rates, different performance, inventiveness, etc. In the majority of the projects, also additional costs occur, e.g. the purchase or creation of multimedia aids or specialised software, leasing the equipment and renting premises. Even having gathered all the data necessary to calculate the costs, the training coordinator with no IT support can verify the history of cooperation with the customer only indicatively, and with a large margin of uncertainty decide what profit margin should be charged on the project and what deadline to set for its implementation. A significant obstacle can also be the simultaneous implementation of many training projects by the company and a lack of a system for their accurate scheduling.

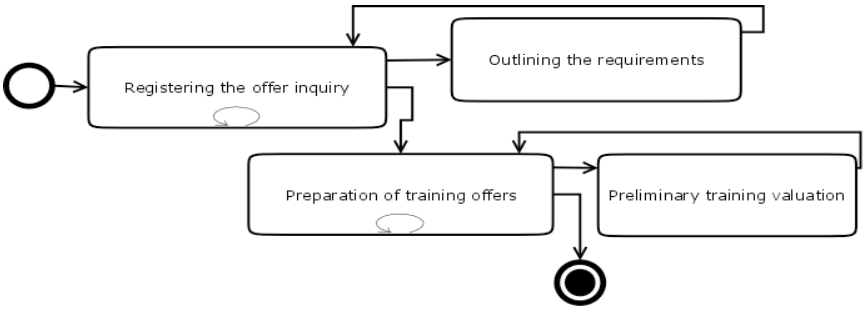

Fig. 3. The structure of the process of managing training offers (own study).

The development of a single training offer takes the company from a few to a dozen of staff-hours. This activity often requires the involvement of the key employees of the training institution, not only the persons responsible for the acquisition of orders and contact with the customer. This results directly in need of IT assistance in the training offer managing process.

The sub-process of recording requests for quotation (see fig. 3) should collect all the customer's requirements in an intelligent way. It means that even in the case when the customer does not have a precise vision of the training needed, an interactive registration form should help them to specify their needs. In the authors' opinion, the basis for the dialogue with the customer during the collection of requirements can be provided by mechanisms used by AI experts on developing computer programs intended for conversation between the human and the computer using a natural language (chatterbot). The most known example of such software is currently the ALICE Bot [13]. Furthermore, on developing the code, that will decide on the completeness and consistency of the order, the use of a neural network is recommended. Having learned, it will enable the creation of requests for quotation taking into account the traditions and the priorities of the training company.

The development of the sub-process of recording the requests for quotation consists in that an algorithm for sketching the customer requirements should be developed, that would simply record the key parameters of the requested training (possibly by making it possible to attach text and graphic files to the registration form: photographs, charts, schematics, notes, and sketches). This sub-process shall require the registration of the customer's representative as the system user. This way, each customer shall have access to their requests for quotation presented in a current, structured form. At the same time, it shall be easy to organise the customer's access to their previous inquiries. Therefore, the time necessary for acquiring the preliminary requirements from the customer can be reduced to the necessary minimum.

The IT system should immediately inform the responsible employee of the training company on the event of receiving a new request for quotation. The key idea for the development of a processing mechanism for such event is that the mechanism of context-based comments run by the system is to enable the responsible employee to immediately communicate with the customer in writing regarding the request and to enable the customer's representative to respond to the employee's communication quickly. The algorithm for sketching the requirements should be used for all requests arriving at the training company. It will enable to save time devoted to the precise formulation of requests for quotation created via the IT system.

After registration, properly developed requests can be easily divided into sub-tasks and distributed to the relevant staff or training service providers. Thanks to the suppliers' participation in generating the quotations, the entire quoting process can be fully automated. The IT system should ensure that the information on the need to create a quotation will be received only by concerned persons. This way, the commercial confidentiality of running negotiations shall also be preserved.

The introduction of a uniform algorithm for the valuation of training in the company shall additionally structure the circulation of documents. Each contractor shall receive the part of the training project documentation corresponding to their competence. With a set of appropriate forms, the valuations will be much more precise and contain the necessary data. They will also allow for the reduction of time spent on the valuation. It will be possible to

submit quotations developed by the training company employees and by external suppliers to automatic analysis with the automatic quotation analysis algorithm that shall verify the baseline variables and shall choose the best bid according to the adopted criterion. The given algorithm constitutes the authors' know-how of the IT system structure.

3.2. The process of preparing the training contracts

Each training company should have specimens of contracts for different variant of cooperation with the customer, prepared by lawyers (contract work, contract work with extended warranty scope, non-disclosure agreement, work contract with a non-disclosure agreement, cooperation agreement, the order template, and other such documents). The selection of the type of the contract for the given training order depends on many factors. For example, if the cooperation agreement has already been signed with the customer then only the order to provide service should be signed for consecutive training. Thus, the knowledge of the actual status of signed documents concerning the given customer is essential for the preparation of new training contracts.

Each time, in order to determine the status of the contracts signed with the customer, it is necessary to check whether the previous agreements have already been signed by the training company, if they have been sent to the customer, if they have been returned to the training company and if their copies have been signed. Only after determining the actual status of the documentation associated with the given customer, it is possible to choose the appropriate specimen for a new contract. On its basis, the provisions concerning the specific training order shall be made precise in cooperation with the customer.

A few decision makers from each party participate in the process of coordinating the contents of the current training contract (see fig. 4). The traditional style of exchanging doubts provided in the comments to a text document or a spreadsheet (MS Office, Open Office, Libre Office), often leads to confusion regarding the currently agreed upon the status of the developed contract. This process, traditionally carried out by exchanging attachments (text documents and spreadsheets) by email, often gets longer and causes delay. Only the determination of the final version of the provisions allows for the contract on cooperation for the given order to be signed formally. In practice, particularly while working with agencies, the time limit for the performance of the training contract often happens to be so short that the training company must start working on the order already before the final agreement is signed.

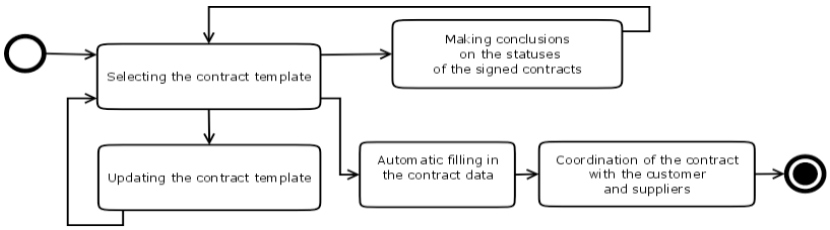

Fig. 4. Key steps of the training agreement development process.

The signed contract must be stored in the documentation of the training provider. After the delivery of the training, the contract together with other documents should be forwarded to the archives. The practice of maintaining documentation does not always favour facilitating access to contracts, especially those archived ones. At the same time, the logic of training business commands returning to the open and completed contracts during the development of further contracts with the same customer or for a similar training with other customers. The development of a computerised knowledge database on the actual status and contents of all the signed training contracts would speed up the process of agreeing upon conditions for the newly created contracts and would improve the quality of documents. The content of that database is the company secret of the training provider.

The automation of the conclusion making process on the statuses of the signed contracts and the selection of the contract specimen that should be the basis for the development of another contract with the specific customer should constitute one of the essential tasks on the implementation of the IT system for training project management. Storing the current electronic copy of each contract in the data warehouse shall eliminate errors connected with the determination of the final form of the newly created contract. Using the collected data should also enable for automatic

filling in the relevant items of the new contract, concerning the customer's attributes and the scope of work to perform. These data must be retrieved from the data warehouse (the central information repository of the training company). With a smaller size company, the warehouse mechanisms may be replaced with database mechanisms.

The key mechanisms of the IT system performing the concluding should enable regular updating of the specimens and provide the function of commenting to all the stakeholders in the project. This will allow to deploy decentralised communication with the stakeholders and to facilitate the creation of the final version of the contract document. The IT system user rights management mechanism should ensure the security of access to the data that might constitute an advantage for the competition.

3.3. The process of scheduling the training projects

On generating the tender, made based on a valuation performed by the employees in charge, the number of staff-hours necessary for the preparation and implementation of the ordered training should be preliminarily estimated. It is also necessary to generate a preliminary schedule of preparatory works and training classes with a safety margin (in terms of time, human, and material resources). The optimal dates for commencement and completion of the training should be estimated, taking into account the involvement of employees in other projects and activities carried out by the training company. Based on the valuations, a larger training project can be divided into steps (modules) which will also require initial valuations and schedules.

Computerisation of training projects scheduling (see fig. 5) requires that the training provider introduce significant changes aimed at:

- structuring the work allocation,
- optimisation of task distribution,
- specification of the time limit calculation algorithm,
- optimising the use of human resources of the company.

The introduction of the changes in the process shall allow for increased effectiveness of the training company, thus - for a significant increase in its income.

Following the signed contracts, the training business management system should enable generating a list of precisely formulated tasks to be performed. The whole training project should be divided into sub-projects, and then to the so-called scenarios. The scenarios should constitute a complete whole from the customer's business point of view. So prepared task units should be ranked for their business value in such a way as to be able to provide the customer with the key values for their business in case of possible delay in the training projects. The introduction of such classification of tasks shall allow for a justified division of the training project into modules. This way it will be possible to actively manage the implementation of the project and report possible problems and changes on a current basis.

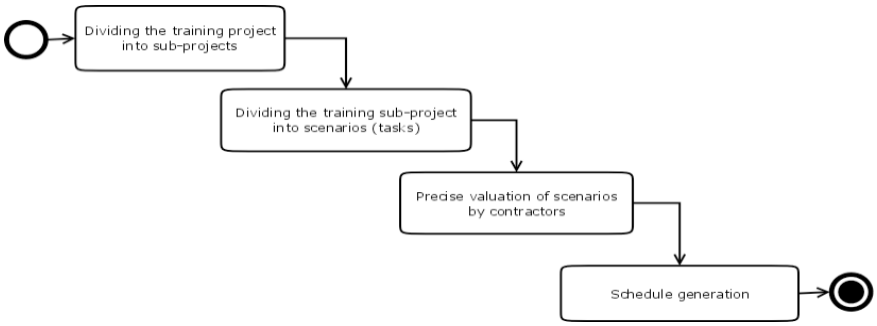

Fig. 5. Key steps of the training scheduling process (own study).

According to the proposed scheduling process structure, the requirements concerning each scenario shall be evaluated before its implementation by the contractors (training company employees) in the units of "difficulty" of the tasks included therein. The application of such an approach will allow estimating, based on a weekly average,

what percentage of the scenario is going to be implemented within the planned period. Using historical statistics concerning the performance of tasks valued before by the same contractors, the system will be able to verify the latest valuation by the contractors and to calculate the calendar time for the performance of tasks by the given individual training team. The implementation of the authors' method of scenarios valuation in connection with historical statistics will allow for more realistic scheduling of tasks, thus for the automated monitoring of the contractors' efficiency and increased customer satisfaction.

A significant change in the approach to scheduling should also be personal changes in teams. In Polish training companies, there aren't usually clear divisions into teams - all employees perform all tasks, which results in a very high overhead for the introduction of new employees. The concept of the authors' IT management system provides for the division of the training company into interdisciplinary mini-teams with the internal specialisation of employees. Such teams will have assigned tasks. Then, in case of exceptions, the system will be able to redirect the task to an alternative team for the former to be solved with the least cost in the prescribed time limit.

3.4. Reporting and training settlement process

The changes in personnel and organisation, unavoidable during the implementation of the project, will create situations in the training company where the process of settlement of training projects with customers and suppliers may cause problems. For example, it happens in practice that due to the errors in the worked hours registers it is difficult to establish how many hours each employee has worked during the given training project. Furthermore, the settlement with some training providers may be done based on the amount agreed upon for the project and based on actually worked hours with others. The analysis of the calendars of the employees implementing the training may of course help in establishing the scope of the performed works, but it is a time-consuming process. The computerisation of the sub-process of reporting the progress of works on the training project will significantly contribute to the increased work discipline and will reduce the number of errors made during the settlement of the training. The reporting can be associated with the training scheduling, which will result in the facilitation of monitoring progress and delays in the training provided.

Determining the scope of work carried out (that should be charged) should be followed by the sub-process of the work acceptance (see fig. 6). It requires the consent of both parties: first, the work is accepted by the training company from its employees and training suppliers, and then - by the end customer (training ordering party). In practice, this two-step business sub-process often results in misunderstanding concerning the status of the ordered service: has it already been finally accepted, or not? The acceptance of work should also include the development of acceptance protocols based on which invoices will be issued. At the same time the task of the settlement of the training should be completed, followed by automatic analysis of payment for the training in the system.

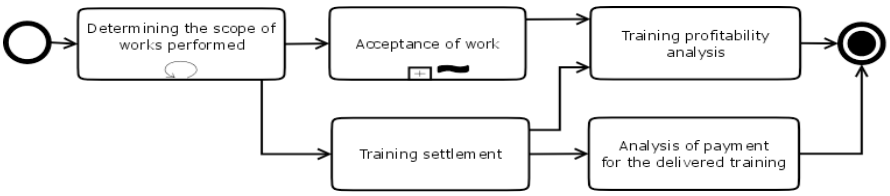

Fig. 6. Reporting and training settlement process (own study).

The documents generated within the reporting and training settlement should be unambiguously associated with the specific training projects, which will facilitate the analysis of those projects profitability. On meeting the given requirement, the management of the training company shall have immediate information indicating that some of the implemented training incur a loss. It is not the only advantage of the forced association of all documents developed by the management system with specific training. It will also enable to increase the effectiveness of project management via the facilitation of chronological analysis of economic indicators for cooperation with each specific customer. Such functionality becomes possible for implementation within the described system. It does not require

very intensive labour from the designer, because it is only about generating relevant reports based on the data already included in the database.

Problems in managing a training company can be caused by impeded analysis of payments. The training company management staff should receive updated information on whether the amounts due from the customers have been credited to the banking account of the training company and if the money transfers to the suppliers have been carried out. This information is related to the state of the banking accounts indicated in the order contracts and the issued invoices; thus the knowledge may be distributed. Such knowledge is centralised in one place only, in the accounting program (at the training company accountant), but, for safety reasons, it must not be connected to the Internet and cannot share the information about the status of the accounts to third parties for analysis. Also, the settlements are often not carried out by the accountants on the current basis (accounting is often provided via outsourcing - by an external entity, and not by own staff).

The training settlement process can be substantially automated by reporting the worked hours by contractors and suppliers, which will allow for automatic charging the amounts for the performance of work. Marking the tasks as "accepted" by the customer in the IT system will allow the training company and the suppliers draw conclusions on the possibility to settle the given order.

By storing the information on the requirements concerning the given customer's order in the system, all stakeholder will have the same common knowledge about the fulfilment of the requirements of the training project. This will allow them to verify the completeness of the performed work better. Automatic counting of the worked hours and the performed task enables to determine the remuneration payable to the contractor (supplier). Automatic reminders on issuing the invoice and automatic monitoring of the fulfilled obligations and receivables will allow saving time devoted to manual management of those areas. With the function of issuing the invoices by the system and the possibility to record cost invoices for the given project, it will be possible to analyse the profitability of each order - this will enable better estimation of future tenders for each customer.

4. The proposed structure of the training management system

Business objects, resulting from the dynamics analysis have been presented in figure 7 in UML notation. It can be seen that the customer's business activity should focus on the precise specification of requirements for the ordered training, and the activity of the training company - on making offers, development, reporting, and settlement of agreements and on scheduling the training project.

Following the process and object analysis carried out, the structure of the module structure of a training management system can be suggested to the developers [14, 15, 16, 17, 18, 19, 20, 21]. First of all, the following system modules should be considered:

- • The request for quotation module;
- • The requirement sketching and specifying module;
- • The tender generation module;
- • The module supporting the development of training contracts;
- • The training scheduling module;
- • The training project progress monitoring module;
- • The reporting and training settlement module.

The above list of modules does not include proposals for obvious elements of each IT system, e.g. for users' management system, documents archiving, or the access to the data repository [22, 23, 24, 25]. Due to the high dynamics of the training business, those modules should include the mobile access module. Enabling the process participants to access the documentation from mobile devices may create a tool for building a competitive advantage in the training services market.

Apart from the listed modules, depending on the business model of the training institution, a module for aiding the cooperation with external suppliers and partners can also be useful.

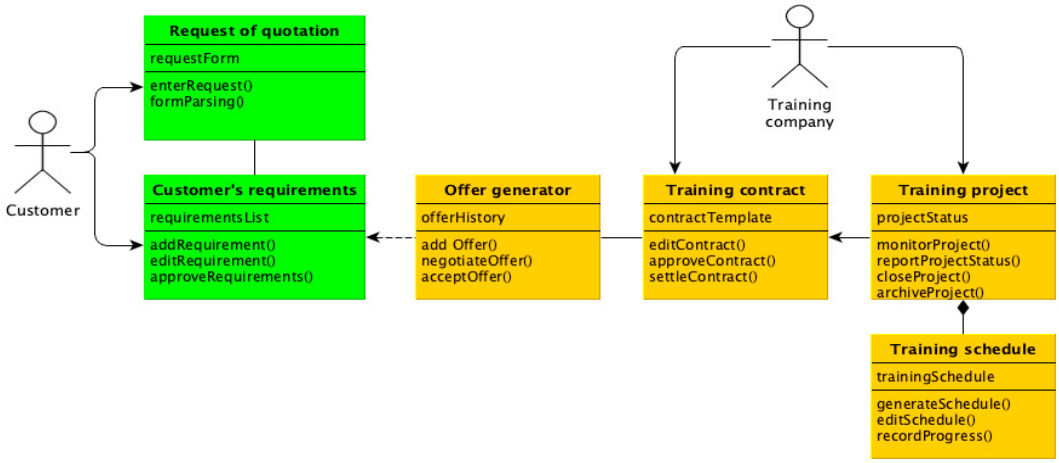

Fig. 7. Business objects of the training company management system (own study).

In conclusion, it is worth to point out that the herein described concept of construction of a training management system has been based on the authors' experience with the intra-academic project of a system for the evaluation of modular effects of education at the Technical University of Koszalin. An interested reader can make themselves familiar with the description of that project studying four of our articles [26, 27, 28, 29, 30, 31, 32]. The solutions used in that system constitute prototypes for some solutions for the management system described herein.

5. Summary

The article presents the assumptions, structured by the authors, for designing an IT system for training company management. The authors expect that the pre-project conceptual business processes modelling in connection with the conceptual and object analysis of the problem domain have made it possible to propose a coherent and logical model of an IT system. The model has been verified in selected fragments during the organisation of training at the Technical University of Koszalin, comprising the use of the academic e-learning platform. The implementation of the concept of development of the system in the form of agreement of a non-exclusive license is the indicator of the success of the research carried out. It also brought tangible financial benefits both for the implementing company and for the Technical University of Koszalin.

Finally, we would like to turn the reader's attention to the fact that the authors have used a three-step scheme for thinking on domain knowledge: from the conceptual model of the domain, through process models and to the conceptual model of the IT system structure. This scheme can be of methodological significance - provide a pattern to follow during the analysis of other domains that should be thoroughly analysed before commissioning a dedicated IT project for them.

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